

AnyCam 3.4.0-rc1.0 Audit Report

Release audit - 04 October 2026 - detection zones - tag 3.4.0-rc1.0

Scope: phase 3.4.0 of the build plan: C17, detection zones, built to CrystalHeeler's 12 requirements (2026-10-02) and the 23 answers approved on 2026-10-04, both in docs/Detection_Zones_Plan.md. Built in the untested run on top of 3.3.0-rc1.0. This build is not field-tested.

1. The answers, as built

#	Answer	Built
1	Close at the first point (12 px) or Enter; 3 points	Yes
2	Zone list; rename; delete with confirm; right-click or long press removes a point; middle handle adds one; Undo/Backspace	Yes
3	Touch: tap; Finish and Close shape buttons; no zoom in the drawing area	Yes (touch-action: none)
4	Overlaps: each zone on its own cells	Yes
5	Sensitivity in the window and the cog panel; Off masks	Yes
6	Crossing lines refused at close	Yes, page and server
7	Name asked at close; Zone N suggested; unique; 40 characters	Yes
8	128 x 96 with zones; warning under 24 cells; re-check whole-picture levels	Grid and warning yes; the re-check needs field data
9	Second picture in the same zone; no one-picture 3% rule in a zone	Yes
10	Light rule on the whole picture only	Yes
11	5 s comparison, two pictures in a row	Yes
12	Night boost on each zone	Yes
13	Zones only: outside not checked	Yes
14	Tuning line and cog reading per zone and outside	Yes
15	Zone in the log and the Storage tab, not the file name	Yes (/data/recording_zones.json, newest 2,000)
16	Show zones; recording zone marked	Yes
17	Zones in /data/motion.json	Yes
18	Pause for a still; snapshot cameras draw on their picture	Yes
19	Picture coordinates	Yes
20	PTZ: the cog panel says zones belong to one view	Yes (for every camera)
21	Zones button; Edit zones in the cog panel	Yes

#	Answer	Built
22	Click outside starts a zone; click inside selects; + New zone; double click stops drawing	Yes
23	Done saves; Cancel confirms; Esc stops drawing only	Yes

2. Detection design

- `anycam_zones.py` holds the geometry (crossing test, point in polygon, cells), the checks of the page's input, and the judging; it imports no other AnyCam file.
- `_motion_cells` returns one flag per changed cell next to the whole-picture share and spread. Each zone counts its own flags; outside counts the cells in no closed zone. A zone's cells are built once per change, not per picture.
- The area that passed by the widest margin over its threshold is the trigger; its name goes to the confirmation rule, the second look on the snapshot path, the log and the recording.
- A change of zones starts the detector again (the grid follows the zones); the recording buffer keeps running.
- The page's own requirement 5 (a click goes on with an open zone) and answer 22 (a click outside starts a new zone) meet like this: with an open zone selected, a click continues it; otherwise a click outside starts a new zone.

3. Also fixed

A WebRTC or RTSP-over-WebSocket camera has status 'info'. `openFocus` opened only cameras with status 'ready', so in 3.2.0-rc1.0 those cameras played in the card but did not open Enhanced View. `openFocus` now also opens them.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 16 files; build inputs checked online.
Python integration	PASS	532 of 532; 19 new in section AE.
JavaScript behaviour	PASS	158 of 158; 19 new in section C17: geometry, the letterbox, clicks, closing, crossing, Esc, resume, Undo, 6 zones, small zones, Off, Done, Cancel, double click.
Undefined names	PASS	16 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order). The image was not built. The drawing window was not seen in a browser: the page tests run its code against stand-in elements, so its look, the drag on a touch screen and Pause on a live video are not checked.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.3.0-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	16 files.
1.4 Async	File writes	PASS	motion.json and recording_zones.json are written in a thread.
1.6 Function design	New file	PASS	Pure functions in anycam_zones.py, testable without the camera state.
1.18 Testing	New tests	PASS	19 server and 19 page checks.
2.x JavaScript	Events	PASS	Pointer events with capture; handlers removed when the window closes.
4.x CSS	Small screens	PASS	The zone panel becomes a bottom sheet under 700 px; 100dvh used.
6.1 Security	Input at the boundary	PASS	Zones validated: count, names, points 0 to 1, levels, crossing lines; zone names escaped wherever the page shows them.
Rule 9 (privacy)	New text	PASS	Tests use made-up names and addresses only.

Known issues carried forward

- Answer 8: the whole-picture levels at 128 x 96 are not checked yet; a camera with zones may need a new whole-picture setting.
- Zones belong to one view of a PTZ camera.
- **B6**: the Microseven waits until CrystalHeeler unlocks it.
- **C21**: the classic engine stays as the fallback.

Acceptance (field test)

- Test system B, ch4: a zone around the far garage door; the door opening records, the still scene does not.
- The tuning line shows the zone's peak; the cog panel lists the zone.
- Drawing works with a mouse on a computer and with a finger on a phone.
- The recording's entry in the Storage tab names the zone.

Files changed

File	Change
anycam_zones.py	New.
anycam_motion.py	Grid, cells, zone judging, slow look, tuning, the zones endpoint, storage.
anycam_storage.py	The zone of each recording.

File	Change
verify_release.py	The comparison contract moved to _motion_cells.
page_script.py, anycam_page.py	The drawing window, Show zones, the cog panel's zones; openFocus for WebRTC and WS-RTSP cameras.
camera_discovery.py, anycam_modules.py, Dockerfile	Route; the new file; version 3.4.0-rc1.0.
tests/	Sections AE and C17.
CHANGELOG.md, config.yaml, docs	3.4.0-rc1.0; zones plan status; build plan.